Set 4A: BJT DC Bias Analysis

ECEN 325 — Lecture 4 (Bipolar Junction Transistors) practice set

What this set trains. Finding the DC collector currents of BJT circuits and checking that every transistor is really active. Assume every transistor active with negligible base current, walk through the circuit node by node (dividers, V_{BE}, KCL), then verify the assumptions. $\beta = 100$ throughout. Problems 1–6 are easy, 7–12 medium, 13–18 hard. Try each problem before looking at its solution.

The Lecture-4 pipeline (use it for every problem in this lecture)

Tools — this is all the theory you need (plus the Lecture-1–3 tools):

T1 *Active BJT at DC:* npn $V_{BE} = 0.7\,\text{V}$, pnp $V_{EB} = 0.7\,\text{V}$; $I_C = \beta I_B$, $I_E = (\beta+1)I_B$, so $I_C \approx I_E$ and, when β is large, $I_B \approx 0$. Active needs $|V_{CE}| \geq 0.2\,\text{V}$.

T2 *Current mirror:* transistors with the same base node and the same emitter node have the same V_{BE}, hence the same I_C.

T3 *Small-signal parameters:* $r_e = \frac{V_T}{I_C}$ ($V_T = 25\,\text{mV}$), $\alpha = \frac{\beta}{\beta+1}$, $r_o = \infty$. For AC, DC supplies are ground and "∞" capacitors are shorts.

T4 *Node impedances and gains:* $Z_{base} = (\beta+1)(r_e + Z_E)$, $Z_{emitter} = r_e + \frac{Z_B}{\beta+1}$, $Z_{collector} = \infty$; CE $\frac{v_c}{v_b} = -\alpha\frac{Z_C}{r_e+Z_E}$, CB $\frac{v_c}{v_e} = \alpha\frac{Z_C}{Z_{emitter}}$, CC $\frac{v_e}{v_b} = \frac{Z_E}{r_e+Z_E}$. Z_C, Z_E, Z_B are everything else hanging on that node (including the input resistance of the next stage).

Steps:

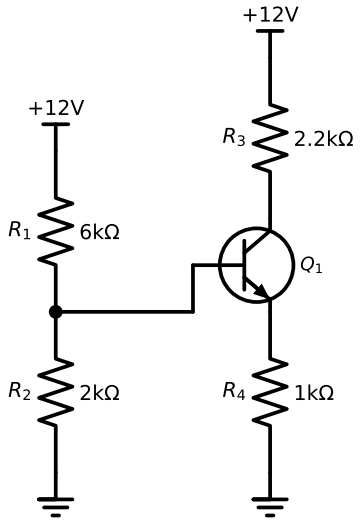
1. *DC:* assume every transistor active and $I_B \approx 0$; walk through the circuit node by node (dividers, V_{BE}, KCL for I_E, $I_C \approx I_E$, collector voltages); then *verify*: $I_B \ll$ bias currents and $|V_{CE}| \geq 0.2\,\text{V}$.
2. *AC:* r_e of every transistor; input resistance of every stage from the last one backwards; multiply the node-to-node gains along the signal path; then standard form, Bode plot and output as in Lecture 1.
3. *Design:* run the same relations backwards, choosing the free quantities with the stated design rules, then check the result by analysing the finished circuit.

Problems

Easy problems

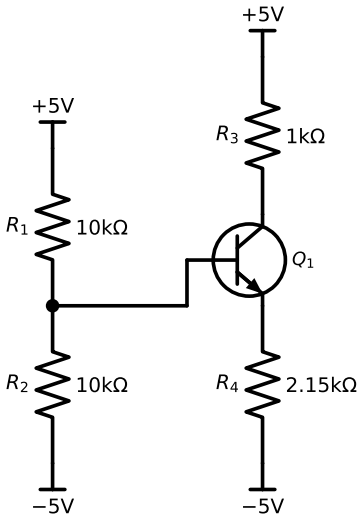
Problem 4A.1 **Easy**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



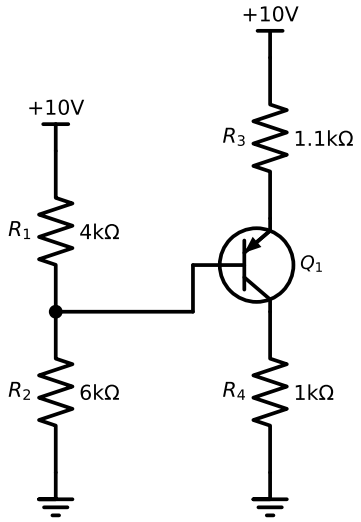
Problem 4A.2 **Easy**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



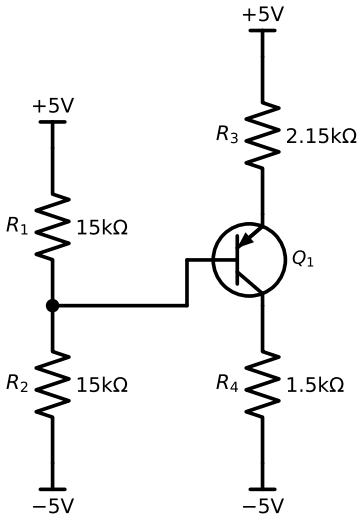
Problem 4A.3 **Easy**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



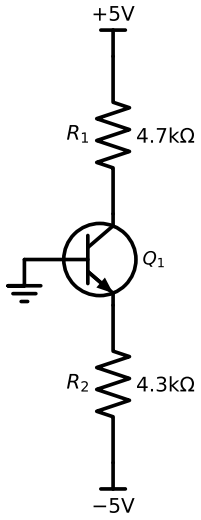
Problem 4A.4 **Easy**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



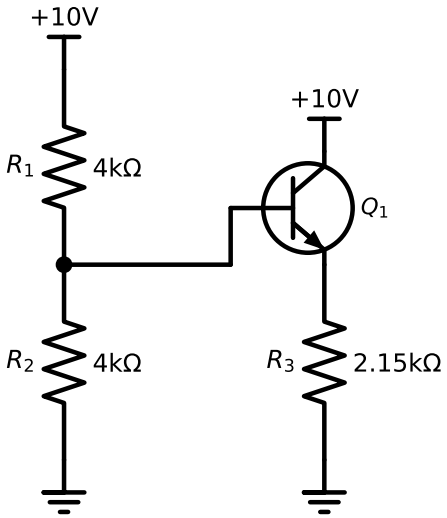
Problem 4A.5 **Easy**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



Problem 4A.6 **Easy**

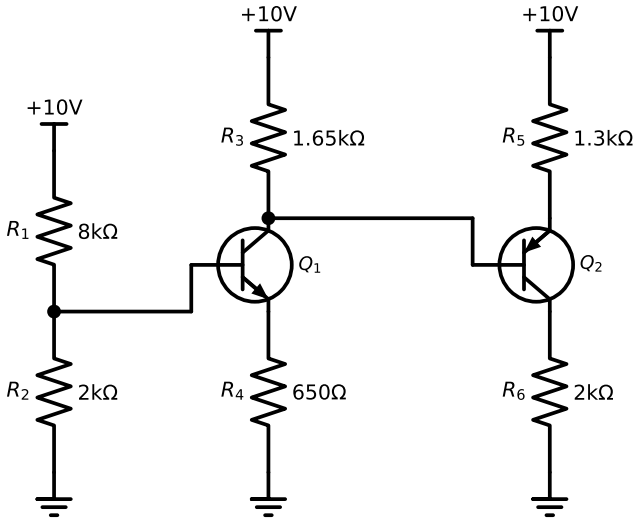
For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



Medium problems

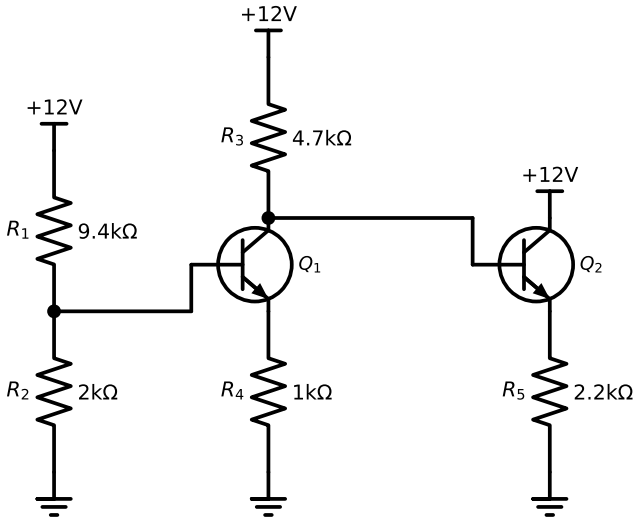
Problem 4A.7 — **Medium**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



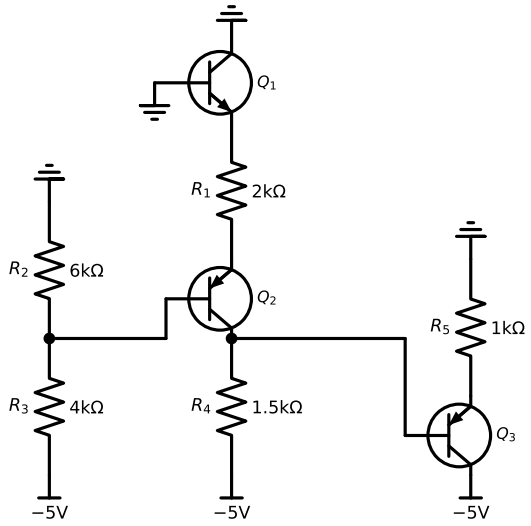
Problem 4A.8 — **Medium**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



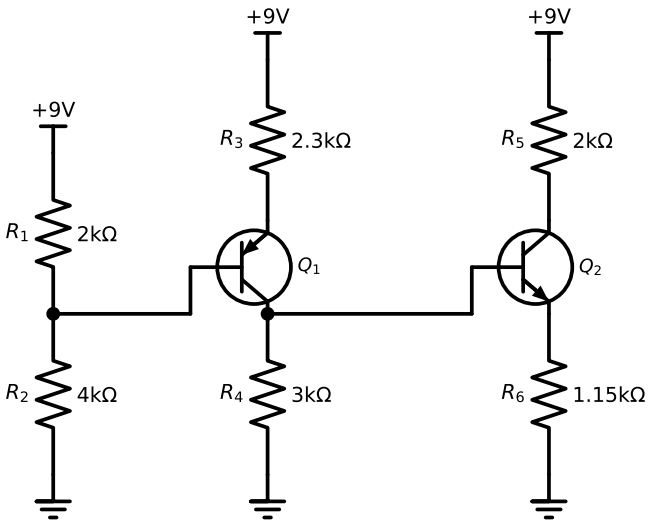
Problem 4A.9 **Medium**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



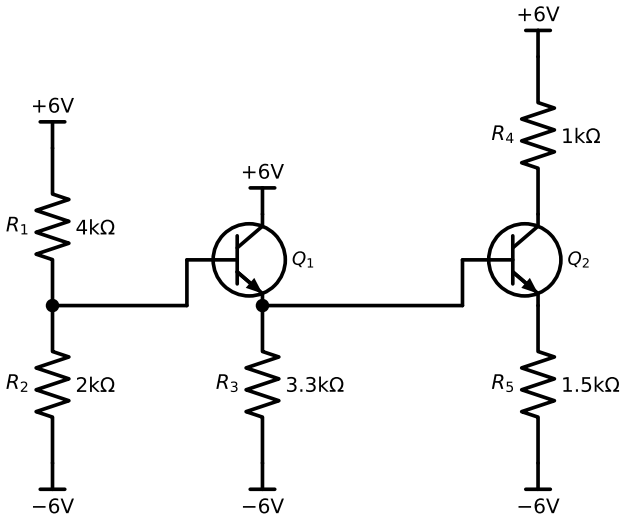
Problem 4A.10 **Medium**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



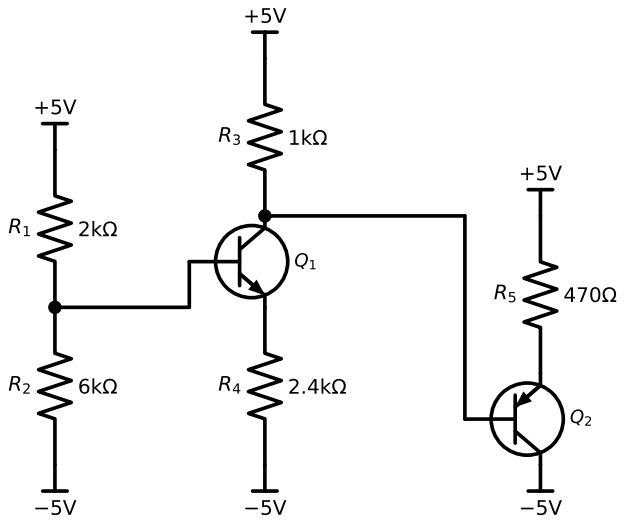
Problem 4A.11 **Medium**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



Problem 4A.12 **Medium**

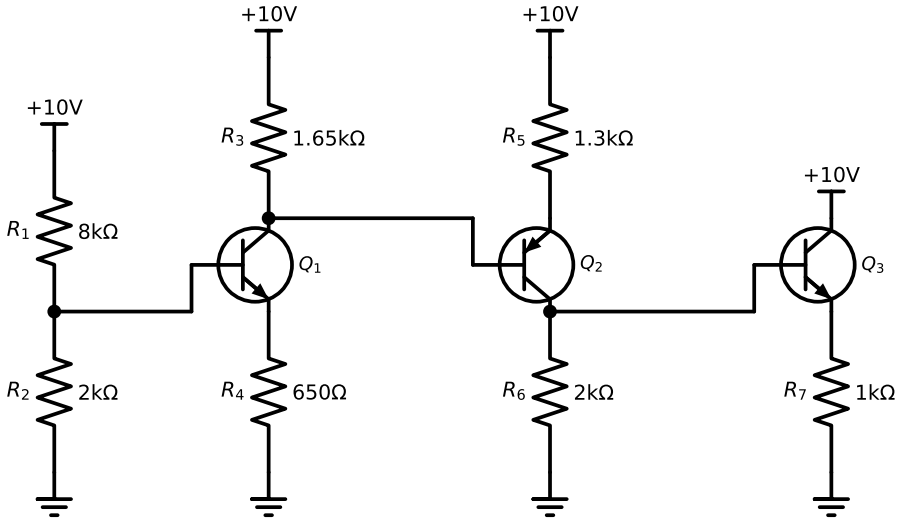
For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



Hard problems

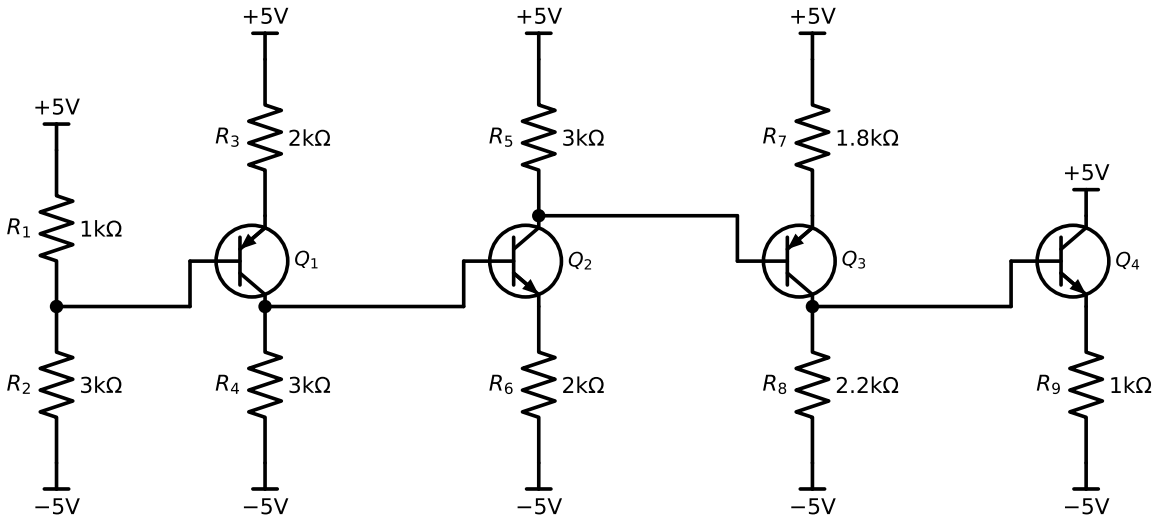
Problem 4A.13 — Hard

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



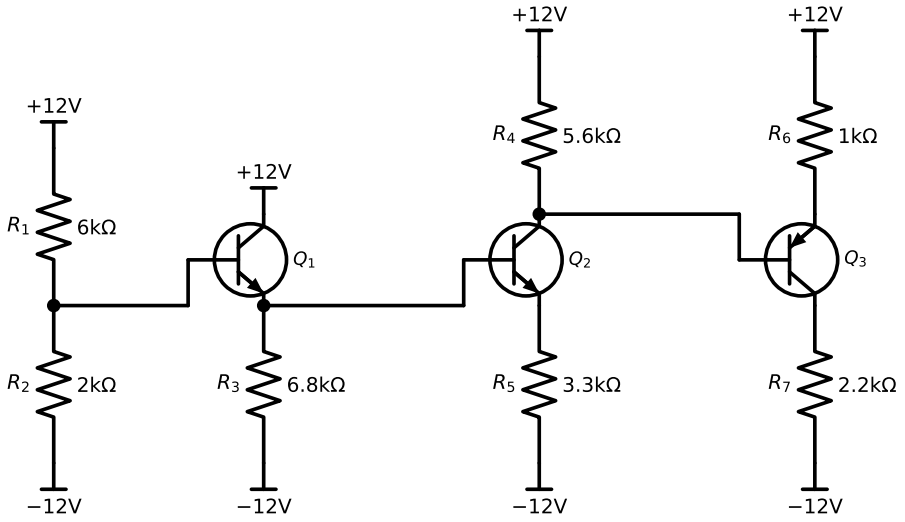
Problem 4A.14 — Hard

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



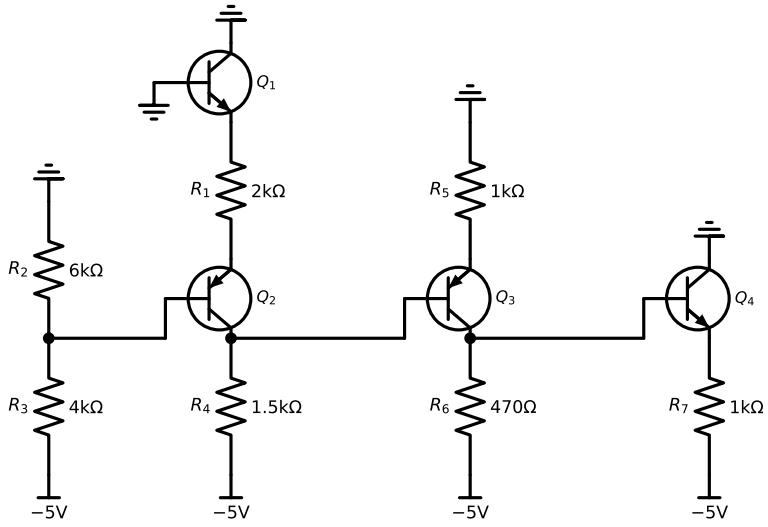
Problem 4A.15 — Hard

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



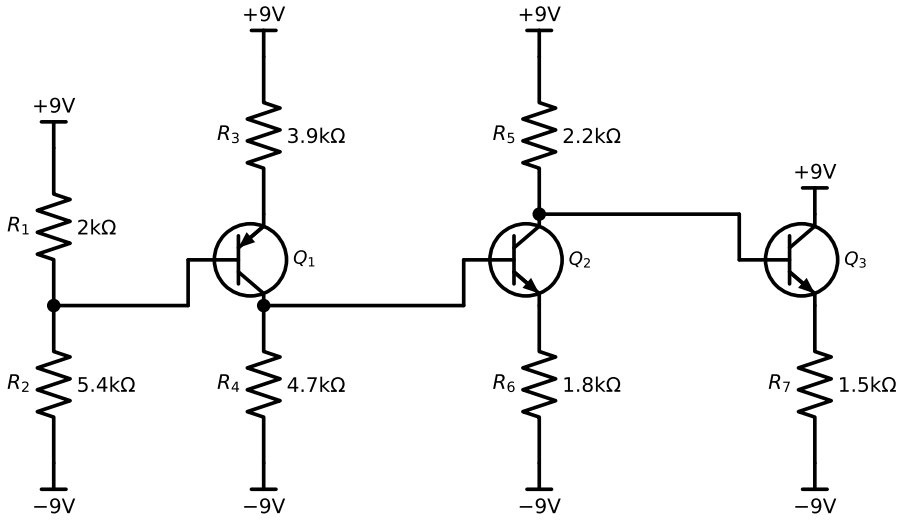
Problem 4A.16 — Hard

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



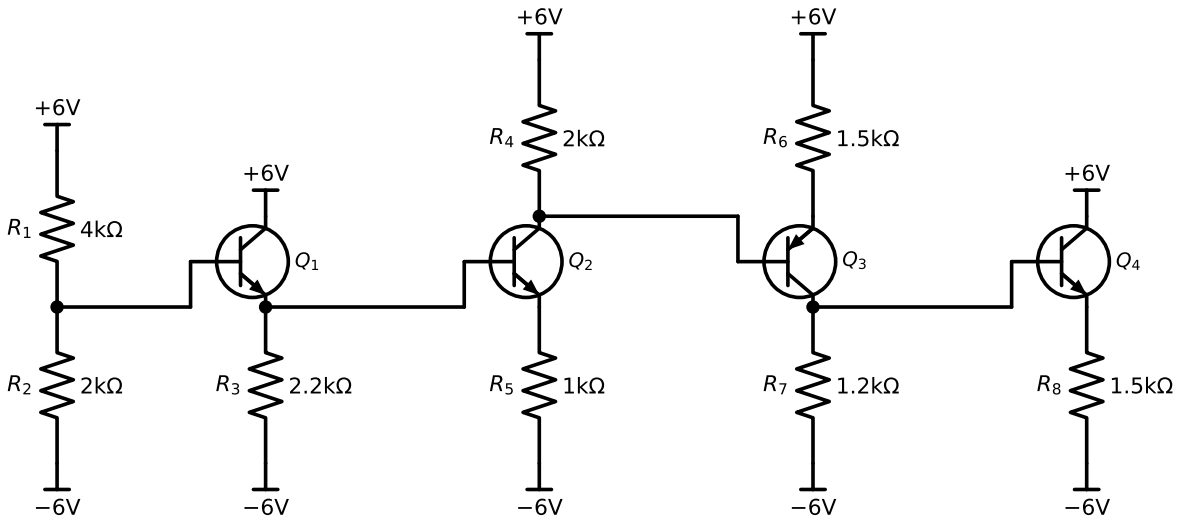
Problem 4A.17 — Hard

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



Problem 4A.18 — Hard

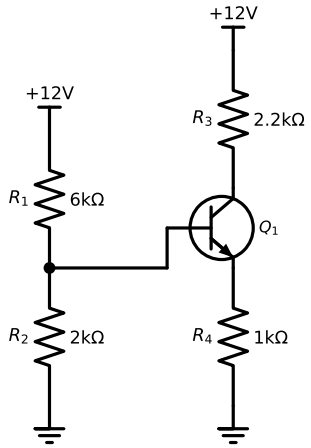
For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



Solutions

Every solution follows the steps of the Lecture-4 pipeline card.

Solution 4A.1 — Easy



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. No current flows into the base (I_B is negligible), so V_{B1} is set by the divider R_1–R_2 between 12 and 0 (Lecture-1 tool):
$$V_{B1} = 0 + (12 - (0))\frac{2\,\text{k}}{6\,\text{k} + 2\,\text{k}} = 3\,\text{V}$$
2. Q_1 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E1} = V_{B1} - 0.7 = 2.3\,\text{V}$.
3. KCL at the emitter of Q_1 (base current neglected):
$$I_{E1} = \frac{V_{E1} - 0}{R_4} = 2.3\,\text{mA}, \qquad I_{C1} \approx I_{E1}$$
4. Q_1 draws $I_{C1} = 2.3\,\text{mA}$ from V_{C1} through R_3: $V_{C1} = 12 - R_3\,(I_{C1}) = 12 - 2.2\,\text{k} \times 2.3\,\text{mA} = 6.94\,\text{V}$.

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 23\,\mu\text{A} \ll$ the 1.5 mA flowing in the base network ✓
- $V_{CE1} = V_{C1} - V_{E1} = 6.94 - (2.3) = 4.64\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓

Step 4 — How good was $I_B \approx 0$?

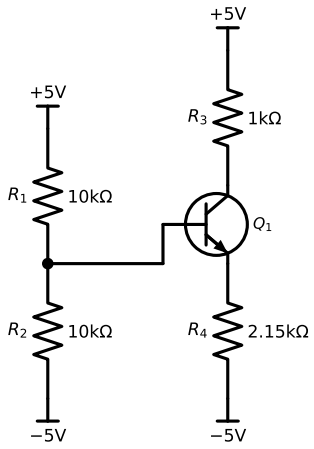
If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	2.3 mA	2.24 mA	2.5%

Answer

$I_{C1} = 2.3\,\text{mA}$
all transistors active, all base currents negligible.

Solution 4A.2 **Easy**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. No current flows into the base (I_B is negligible), so V_{B1} is set by the divider R_1–R_2 between 5 and (-5) (Lecture-1 tool):
$$V_{B1} = -5 + (5 - (-5))\frac{10\,\text{k}}{10\,\text{k} + 10\,\text{k}} = 0\,\text{V}$$
2. Q_1 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E1} = V_{B1} - 0.7 = -0.7\,\text{V}$.
3. KCL at the emitter of Q_1 (base current neglected):
$$I_{E1} = \frac{V_{E1} - (-5)}{R_4} = 2\,\text{mA}, \qquad I_{C1} \approx I_{E1}$$
4. Q_1 draws $I_{C1} = 2\,\text{mA}$ from V_{C1} through R_3: $V_{C1} = 5 - R_3\,(I_{C1}) = 5 - 1\,\text{k} \times 2\,\text{mA} = 3\,\text{V}$.

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 20\,\mu\text{A} \ll$ the $500\,\mu\text{A}$ flowing in the base network ✓
- $V_{CE1} = V_{C1} - V_{E1} = 3 - (-0.7) = 3.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓

Step 4 — How good was $I_B \approx 0$?

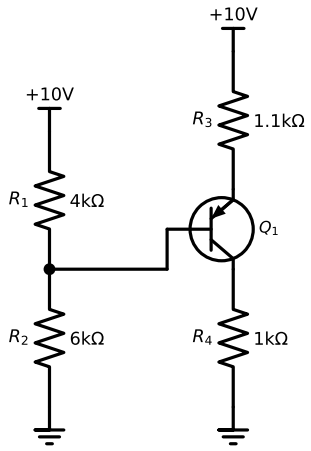
If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	2 mA	1.94 mA	3.3%

Answer

$I_{C1} = 2\,\mathrm{mA}$
all transistors active, all base currents negligible.

Solution 4A.3 **Easy**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. No current flows into the base (I_B is negligible), so V_{B1} is set by the divider R_1–R_2 between 10 and 0 (Lecture-1 tool):
$$V_{B1} = 0 + (10 - (0))\frac{6\,\text{k}}{4\,\text{k} + 6\,\text{k}} = 6\,\text{V}$$
2. Q_1 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{E1} = V_{B1} + 0.7 = 6.7\,\text{V}$.
3. KCL at the emitter of Q_1 (base current neglected):
$$I_{E1} = \frac{10 - V_{E1}}{R_3} = 3\,\text{mA}, \qquad I_{C1} \approx I_{E1}$$
4. The current $I_{C1} = 3\,\text{mA}$ flows from V_{C1} through R_4: $V_{C1} = 0 + R_4\,(I_{C1}) = 0 + 1\,\text{k} \times 3\,\text{mA} = 3\,\text{V}$.

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 30\,\mu\text{A} \ll$ the 1 mA flowing in the base network ✓
- $V_{EC1} = V_{E1} - V_{C1} = 6.7 - (3) = 3.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓

Step 4 — How good was $I_B \approx 0$?

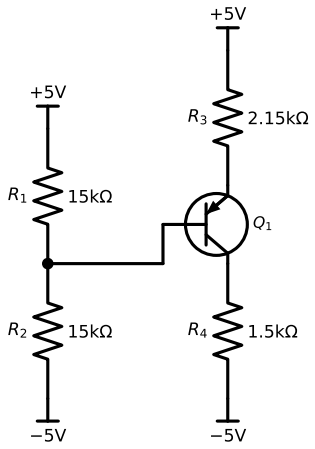
If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	3 mA	2.91 mA	3.2%

Answer

$I_{C1} = 3\,\mathrm{mA}$
all transistors active, all base currents negligible.

Solution 4A.4 **Easy**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. No current flows into the base (I_B is negligible), so V_{B1} is set by the divider R_1–R_2 between 5 and (-5) (Lecture-1 tool):
$$V_{B1} = -5 + (5 - (-5))\frac{15\,\text{k}}{15\,\text{k} + 15\,\text{k}} = 0\,\text{V}$$
2. Q_1 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{E1} = V_{B1} + 0.7 = 0.7\,\text{V}$.
3. KCL at the emitter of Q_1 (base current neglected):
$$I_{E1} = \frac{5 - V_{E1}}{R_3} = 2\,\text{mA}, \qquad I_{C1} \approx I_{E1}$$
4. The current $I_{C1} = 2\,\text{mA}$ flows from V_{C1} through R_4: $V_{C1} = (-5) + R_4\,(I_{C1}) = -5 + 1.5\,\text{k} \times 2\,\text{mA} = -2\,\text{V}$.

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 20\,\mu\text{A} \ll$ the $333\,\mu\text{A}$ flowing in the base network ✓
- $V_{EC1} = V_{E1} - V_{C1} = 0.7 - (-2) = 2.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓

Step 4 — How good was $I_B \approx 0$?

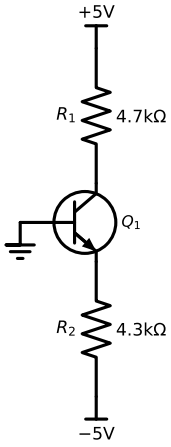
If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	2 mA	1.91 mA	4.5%

Answer

$I_{C1} = 2\,\mathrm{mA}$
all transistors active, all base currents negligible.

Solution 4A.5 **Easy**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. Q_1 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E1} = 0 - 0.7 = -0.7\,\text{V}$.
2. KCL at the emitter of Q_1 (base current neglected):

$$I_{E1} = \frac{V_{E1} - (-5)}{R_2} = 1\,\text{mA}, \qquad I_{C1} \approx I_{E1}$$

3. Q_1 draws $I_{C1} = 1\,\text{mA}$ from V_{C1} through R_1: $V_{C1} = 5 - R_1\,(I_{C1}) = 5 - 4.7\,\text{k} \times 1\,\text{mA} = 0.3\,\text{V}$.

Step 3 — Verify the assumptions

- $V_{CE1} = V_{C1} - V_{E1} = 0.3 - (-0.7) = 1\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓

Step 4 — How good was $I_B \approx 0$?

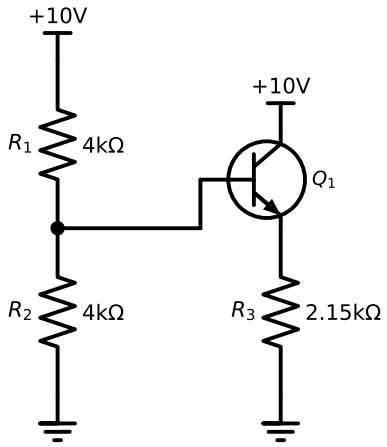
If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	$1\,\text{mA}$	$990\,\mu\text{A}$	1%

Answer

$I_{C1} = 1\,\text{mA}$
all transistors active, all base currents negligible.

Solution 4A.6 **Easy**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. No current flows into the base (I_B is negligible), so V_{B1} is set by the divider R_1–R_2 between 10 and 0 (Lecture-1 tool):
$$V_{B1} = 0 + (10 - (0))\frac{4\,\text{k}}{4\,\text{k} + 4\,\text{k}} = 5\,\text{V}$$
2. Q_1 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E1} = V_{B1} - 0.7 = 4.3\,\text{V}$.
3. KCL at the emitter of Q_1 (base current neglected):
$$I_{E1} = \frac{V_{E1} - 0}{R_3} = 2\,\text{mA}, \qquad I_{C1} \approx I_{E1}$$

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 20\,\mu\text{A} \ll$ the 1.25 mA flowing in the base network ✓
- $V_{CE1} = 10 - V_{E1} = 10 - (4.3) = 5.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓

Step 4 — How good was $I_B \approx 0$?

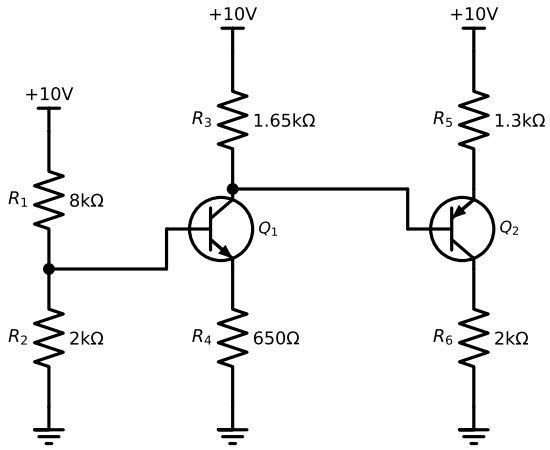
If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	2 mA	1.96 mA	1.9%

Answer

$I_{C1} = 2\,\mathrm{mA}$
all transistors active, all base currents negligible.

Solution 4A.7 **Medium**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. No current flows into the base (I_B is negligible), so V_{B1} is set by the divider R_1–R_2 between 10 and 0 (Lecture-1 tool):
$$V_{B1} = 0 + (10 - (0))\frac{2\,\text{k}}{8\,\text{k} + 2\,\text{k}} = 2\,\text{V}$$
2. Q_1 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E1} = V_{B1} - 0.7 = 1.3\,\text{V}$.
3. KCL at the emitter of Q_1 (base current neglected):
$$I_{E1} = \frac{V_{E1} - 0}{R_4} = 2\,\text{mA}, \qquad I_{C1} \approx I_{E1}$$
4. Q_1 draws $I_{C1} = 2\,\text{mA}$ from V_{C1} through R_3: $V_{C1} = 10 - R_3\,(I_{C1}) = 10 - 1.65\,\text{k} \times 2\,\text{mA} = 6.7\,\text{V}$.
5. Q_2 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{E2} = V_{C1} + 0.7 = 7.4\,\text{V}$.
6. KCL at the emitter of Q_2 (base current neglected):
$$I_{E2} = \frac{10 - V_{E2}}{R_5} = 2\,\text{mA}, \qquad I_{C2} \approx I_{E2}$$
7. The current $I_{C2} = 2\,\text{mA}$ flows from V_{C2} through R_6: $V_{C2} = 0 + R_6\,(I_{C2}) = 0 + 2\,\text{k} \times 2\,\text{mA} = 4\,\text{V}$.

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 20\,\mu\text{A} \ll$ the 1 mA flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 20\,\mu\text{A} \ll$ the 2 mA flowing in the base network ✓
- $V_{CE1} = V_{C1} - V_{E1} = 6.7 - (1.3) = 5.4\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓

- $V_{EC2} = V_{E2} - V_{C2} = 7.4 - (4) = 3.4\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓

Step 4 — How good was $I_B \approx 0$?

If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

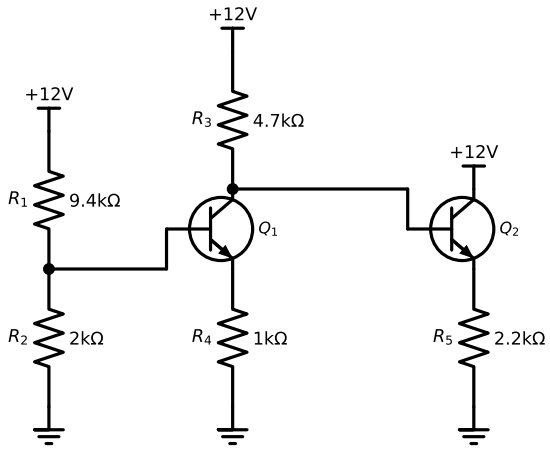
transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	2 mA	1.93 mA	3.5%
Q_2	2 mA	1.87 mA	6.8%

The differences grow along the chain: every base current slightly shifts the collector voltage that biases the next transistor, and $\alpha = \beta/(\beta+1)$ makes each I_C about 1 % smaller than its I_E. A shift is amplified when the next stage has only a small voltage across its emitter resistor. Hand analysis with $I_B \approx 0$ is still the standard first pass.

Answer

$I_{C1} = 2\,\text{mA}, \quad I_{C2} = 2\,\text{mA}$
all transistors active, all base currents negligible.

Solution 4A.8 **Medium**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. No current flows into the base (I_B is negligible), so V_{B1} is set by the divider R_1–R_2 between 12 and 0 (Lecture-1 tool):
$$V_{B1} = 0 + (12 - (0))\frac{2\,\text{k}}{9.4\,\text{k} + 2\,\text{k}} = 2.11\,\text{V}$$
2. Q_1 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E1} = V_{B1} - 0.7 = 1.41\,\text{V}$.
3. KCL at the emitter of Q_1 (base current neglected):
$$I_{E1} = \frac{V_{E1} - 0}{R_4} = 1.41\,\text{mA}, \qquad I_{C1} \approx I_{E1}$$
4. Q_1 draws $I_{C1} = 1.41\,\text{mA}$ from V_{C1} through R_3: $V_{C1} = 12 - R_3\,(I_{C1}) = 12 - 4.7\,\text{k} \times 1.41\,\text{mA} = 5.4\,\text{V}$.
5. Q_2 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E2} = V_{C1} - 0.7 = 4.7\,\text{V}$.
6. KCL at the emitter of Q_2 (base current neglected):
$$I_{E2} = \frac{V_{E2} - 0}{R_5} = 2.13\,\text{mA}, \qquad I_{C2} \approx I_{E2}$$

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 14.1\,\mu\text{A} \ll$ the 1.05 mA flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 21.3\,\mu\text{A} \ll$ the 1.41 mA flowing in the base network ✓
- $V_{CE1} = V_{C1} - V_{E1} = 5.395 - (1.405) = 3.99\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{CE2} = 12 - V_{E2} = 12 - (4.695) = 7.3\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓

Step 4 — How good was $I_B \approx 0$?

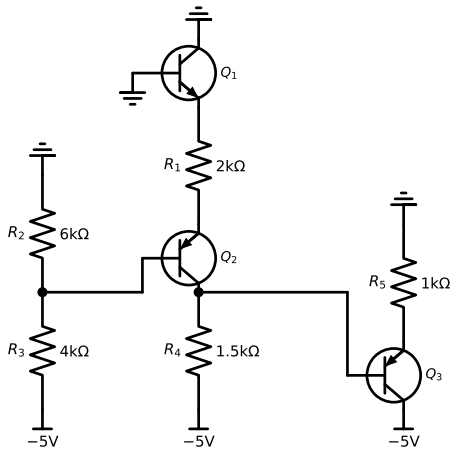
If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	1.41 mA	1.37 mA	2.6%
Q_2	2.13 mA	2.14 mA	−0.48%

Answer

$I_{C1} = 1.41\,\text{mA}, \quad I_{C2} = 2.13\,\text{mA}$
all transistors active, all base currents negligible.

Solution 4A.9 **Medium**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. Q_1 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E1} = 0 - 0.7 = -0.7\,\text{V}$.
2. No current flows into the base (I_B is negligible), so V_{B2} is set by the divider R_2–R_3 between 0 and (-5) (Lecture-1 tool):
$$V_{B2} = -5 + (0 - (-5))\frac{4\,\text{k}}{6\,\text{k} + 4\,\text{k}} = -3\,\text{V}$$
3. Q_2 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{E2} = V_{B2} + 0.7 = -2.3\,\text{V}$.
4. KCL at the emitter of Q_1 (base current neglected):
$$I_{E1} = \frac{V_{E1} - V_{E2}}{R_1} = 800\,\mu\text{A}, \qquad I_{C1} \approx I_{E1}$$
5. KCL at the emitter of Q_2 (base current neglected):
$$I_{E2} = \frac{V_{E1} - V_{E2}}{R_1} = 800\,\mu\text{A}, \qquad I_{C2} \approx I_{E2}$$
6. The current $I_{C2} = 800\,\mu\text{A}$ flows from V_{C2} through R_4: $V_{C2} = (-5) + R_4\,(I_{C2}) = -5 + 1.5\,\text{k} \times 800\,\mu\text{A} = -3.8\,\text{V}$.
7. Q_3 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{E3} = V_{C2} + 0.7 = -3.1\,\text{V}$.
8. KCL at the emitter of Q_3 (base current neglected):
$$I_{E3} = \frac{0 - V_{E3}}{R_5} = 3.1\,\text{mA}, \qquad I_{C3} \approx I_{E3}$$

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 8\,\mu\text{A} \ll$ the 3.1 mA flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 8\,\mu\text{A} \ll$ the 500 μA flowing in the base network ✓
- $I_{B3} = \frac{I_{C3}}{\beta} = 31\,\mu\text{A} \ll$ the 800 μA flowing in the base network ✓
- $V_{CE1} = 0 - V_{E1} = 0 - (-0.7) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{EC2} = V_{E2} - V_{C2} = -2.3 - (-3.8) = 1.5\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓
- $V_{EC3} = V_{E3} - (-5) = -3.1 - (-5) = 1.9\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_3$ is active ✓

Step 4 — How good was $I_B \approx 0$?

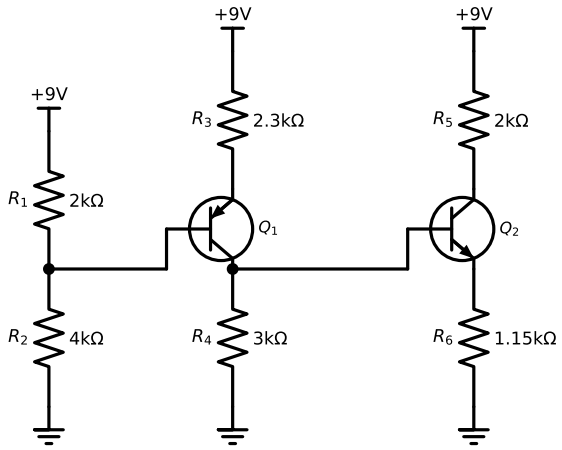
If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	800 μA	783 μA	2.2%
Q_2	800 μA	783 μA	2.2%
Q_3	3.1 mA	3.05 mA	1.7%

Answer

$I_{C1} = 800\,\mu\text{A}, \quad I_{C2} = 800\,\mu\text{A}, \quad I_{C3} = 3.1\,\text{mA}$
all transistors active, all base currents negligible.

Solution 4A.10 **Medium**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. No current flows into the base (I_B is negligible), so V_{B1} is set by the divider R_1–R_2 between 9 and 0 (Lecture-1 tool):
$$V_{B1} = 0 + (9 - (0))\frac{4\,\text{k}}{2\,\text{k} + 4\,\text{k}} = 6\,\text{V}$$
2. Q_1 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{E1} = V_{B1} + 0.7 = 6.7\,\text{V}$.
3. KCL at the emitter of Q_1 (base current neglected):
$$I_{E1} = \frac{9 - V_{E1}}{R_3} = 1\,\text{mA}, \qquad I_{C1} \approx I_{E1}$$
4. The current $I_{C1} = 1\,\text{mA}$ flows from V_{C1} through R_4: $V_{C1} = 0 + R_4\,(I_{C1}) = 0 + 3\,\text{k} \times 1\,\text{mA} = 3\,\text{V}$.
5. Q_2 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E2} = V_{C1} - 0.7 = 2.3\,\text{V}$.
6. KCL at the emitter of Q_2 (base current neglected):
$$I_{E2} = \frac{V_{E2} - 0}{R_6} = 2\,\text{mA}, \qquad I_{C2} \approx I_{E2}$$
7. Q_2 draws $I_{C2} = 2\,\text{mA}$ from V_{C2} through R_5: $V_{C2} = 9 - R_5\,(I_{C2}) = 9 - 2\,\text{k} \times 2\,\text{mA} = 5\,\text{V}$.

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 10\,\mu\text{A} \ll$ the 1.5 mA flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 20\,\mu\text{A} \ll$ the 1 mA flowing in the base network ✓
- $V_{EC1} = V_{E1} - V_{C1} = 6.7 - (3) = 3.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓

- $V_{CE2} = V_{C2} - V_{E2} = 5 - (2.3) = 2.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓

Step 4 — How good was $I_B \approx 0$?

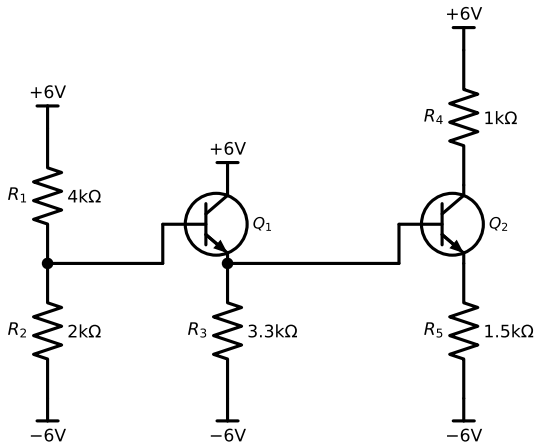
If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	$1\,\text{mA}$	$984\,\mu\text{A}$	1.6%
Q_2	$2\,\text{mA}$	$1.89\,\text{mA}$	5.8%

Answer

$I_{C1} = 1\,\text{mA}, \quad I_{C2} = 2\,\text{mA}$
all transistors active, all base currents negligible.

Solution 4A.11 **Medium**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. No current flows into the base (I_B is negligible), so V_{B1} is set by the divider R_1–R_2 between 6 and (-6) (Lecture-1 tool):
$$V_{B1} = -6 + (6 - (-6))\frac{2\,\text{k}}{4\,\text{k} + 2\,\text{k}} = -2\,\text{V}$$
2. Q_1 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E1} = V_{B1} - 0.7 = -2.7\,\text{V}$.
3. Q_2 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E2} = V_{E1} - 0.7 = -3.4\,\text{V}$.
4. KCL at the emitter of Q_1 (base current neglected):
$$I_{E1} = \frac{V_{E1} - (-6)}{R_3} = 1\,\text{mA}, \qquad I_{C1} \approx I_{E1}$$
5. KCL at the emitter of Q_2 (base current neglected):
$$I_{E2} = \frac{V_{E2} - (-6)}{R_5} = 1.73\,\text{mA}, \qquad I_{C2} \approx I_{E2}$$
6. Q_2 draws $I_{C2} = 1.73\,\text{mA}$ from V_{C2} through R_4: $V_{C2} = 6 - R_4\,(I_{C2}) = 6 - 1\,\text{k} \times 1.73\,\text{mA} = 4.27\,\text{V}$.

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 10\,\mu\text{A} \ll$ the 2 mA flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 17.3\,\mu\text{A} \ll$ the 1 mA flowing in the base network ✓
- $V_{CE1} = 6 - V_{E1} = 6 - (-2.7) = 8.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{CE2} = V_{C2} - V_{E2} = 4.267 - (-3.4) = 7.67\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓

Step 4 — How good was $I_B \approx 0$?

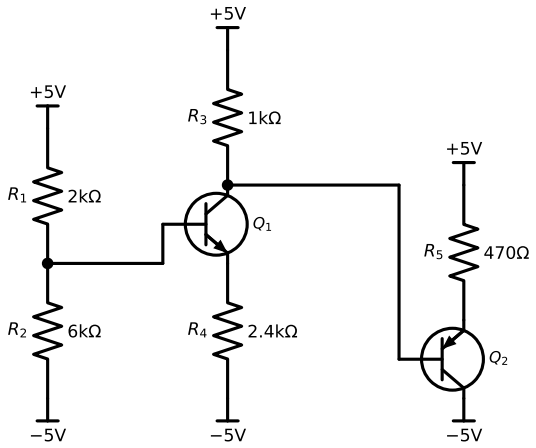
If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	$1\,\mathrm{mA}$	$1\,\mathrm{mA}$	-0.3%
Q_2	$1.73\,\mathrm{mA}$	$1.71\,\mathrm{mA}$	1.5%

Answer

$I_{C1} = 1\,\mathrm{mA}, \quad I_{C2} = 1.73\,\mathrm{mA}$
all transistors active, all base currents negligible.

Solution 4A.12 **Medium**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. No current flows into the base (I_B is negligible), so V_{B1} is set by the divider R_1–R_2 between 5 and (-5) (Lecture-1 tool):
$$V_{B1} = -5 + (5 - (-5))\frac{6\,\text{k}}{2\,\text{k} + 6\,\text{k}} = 2.5\,\text{V}$$
2. Q_1 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E1} = V_{B1} - 0.7 = 1.8\,\text{V}$.
3. KCL at the emitter of Q_1 (base current neglected):
$$I_{E1} = \frac{V_{E1} - (-5)}{R_4} = 2.83\,\text{mA}, \qquad I_{C1} \approx I_{E1}$$
4. Q_1 draws $I_{C1} = 2.83\,\text{mA}$ from V_{C1} through R_3: $V_{C1} = 5 - R_3\,(I_{C1}) = 5 - 1\,\text{k} \times 2.83\,\text{mA} = 2.17\,\text{V}$.
5. Q_2 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{E2} = V_{C1} + 0.7 = 2.87\,\text{V}$.
6. KCL at the emitter of Q_2 (base current neglected):
$$I_{E2} = \frac{5 - V_{E2}}{R_5} = 4.54\,\text{mA}, \qquad I_{C2} \approx I_{E2}$$

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 28.3\,\mu\text{A} \ll$ the 1.25 mA flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 45.4\,\mu\text{A} \ll$ the 2.83 mA flowing in the base network ✓
- $V_{CE1} = V_{C1} - V_{E1} = 2.167 - (1.8) = 0.367\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{EC2} = V_{E2} - (-5) = 2.867 - (-5) = 7.87\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓

Step 4 — How good was $I_B \approx 0$?

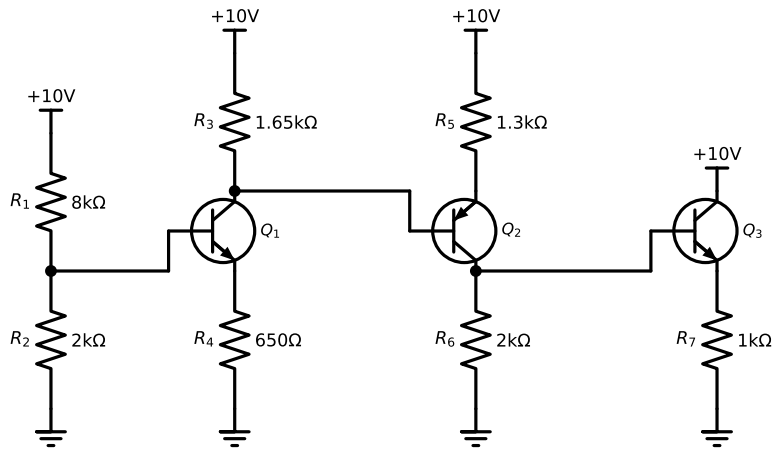
If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	2.83 mA	2.79 mA	1.6%
Q_2	4.54 mA	4.31 mA	5.4%

Answer

$I_{C1} = 2.83\,\text{mA}, \quad I_{C2} = 4.54\,\text{mA}$
all transistors active, all base currents negligible.

Solution 4A.13 **Hard**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. No current flows into the base (I_B is negligible), so V_{B1} is set by the divider R_1–R_2 between 10 and 0 (Lecture-1 tool):
$$V_{B1} = 0 + (10 - (0))\frac{2\,\text{k}}{8\,\text{k} + 2\,\text{k}} = 2\,\text{V}$$
2. Q_1 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E1} = V_{B1} - 0.7 = 1.3\,\text{V}$.
3. KCL at the emitter of Q_1 (base current neglected):
$$I_{E1} = \frac{V_{E1} - 0}{R_4} = 2\,\text{mA}, \qquad I_{C1} \approx I_{E1}$$
4. Q_1 draws $I_{C1} = 2\,\text{mA}$ from V_{C1} through R_3: $V_{C1} = 10 - R_3\,(I_{C1}) = 10 - 1.65\,\text{k} \times 2\,\text{mA} = 6.7\,\text{V}$.
5. Q_2 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{E2} = V_{C1} + 0.7 = 7.4\,\text{V}$.
6. KCL at the emitter of Q_2 (base current neglected):
$$I_{E2} = \frac{10 - V_{E2}}{R_5} = 2\,\text{mA}, \qquad I_{C2} \approx I_{E2}$$
7. The current $I_{C2} = 2\,\text{mA}$ flows from V_{C2} through R_6: $V_{C2} = 0 + R_6\,(I_{C2}) = 0 + 2\,\text{k} \times 2\,\text{mA} = 4\,\text{V}$.
8. Q_3 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E3} = V_{C2} - 0.7 = 3.3\,\text{V}$.
9. KCL at the emitter of Q_3 (base current neglected):
$$I_{E3} = \frac{V_{E3} - 0}{R_7} = 3.3\,\text{mA}, \qquad I_{C3} \approx I_{E3}$$

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 20\,\mu\text{A} \ll$ the 1 mA flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 20\,\mu\text{A} \ll$ the 2 mA flowing in the base network ✓
- $I_{B3} = \frac{I_{C3}}{\beta} = 33\,\mu\text{A} \ll$ the 2 mA flowing in the base network ✓
- $V_{CE1} = V_{C1} - V_{E1} = 6.7 - (1.3) = 5.4\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{EC2} = V_{E2} - V_{C2} = 7.4 - (4) = 3.4\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓
- $V_{CE3} = 10 - V_{E3} = 10 - (3.3) = 6.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_3$ is active ✓

Step 4 — How good was $I_B \approx 0$?

If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

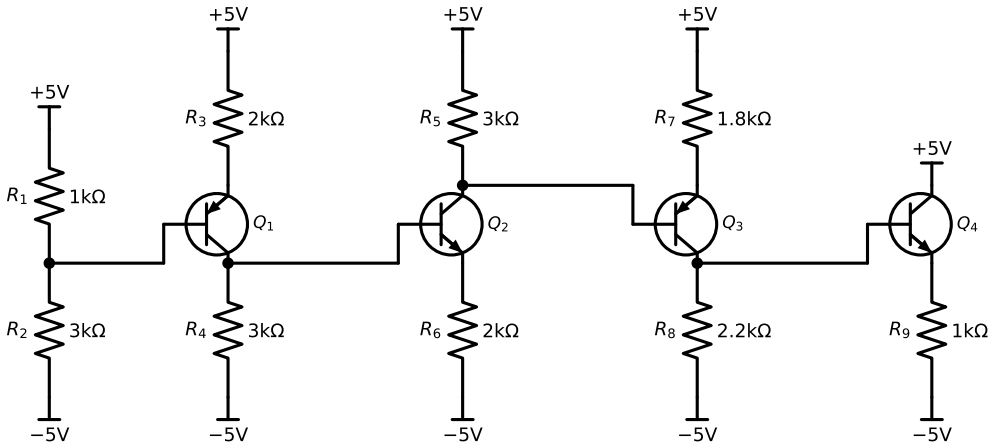
transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	2 mA	1.93 mA	3.5%
Q_2	2 mA	1.87 mA	6.8%
Q_3	3.3 mA	2.96 mA	12%

The differences grow along the chain: every base current slightly shifts the collector voltage that biases the next transistor, and $\alpha = \beta/(\beta+1)$ makes each I_C about 1 % smaller than its I_E. A shift is amplified when the next stage has only a small voltage across its emitter resistor. Hand analysis with $I_B \approx 0$ is still the standard first pass.

Answer

$I_{C1} = 2\,\text{mA}, \quad I_{C2} = 2\,\text{mA}, \quad I_{C3} = 3.3\,\text{mA}$
all transistors active, all base currents negligible.

Solution 4A.14 **Hard**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. No current flows into the base (I_B is negligible), so V_{B1} is set by the divider R_1–R_2 between 5 and (-5) (Lecture-1 tool):
$$V_{B1} = -5 + (5 - (-5))\frac{3\,\text{k}}{1\,\text{k} + 3\,\text{k}} = 2.5\,\text{V}$$
2. Q_1 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{E1} = V_{B1} + 0.7 = 3.2\,\text{V}$.
3. KCL at the emitter of Q_1 (base current neglected):
$$I_{E1} = \frac{5 - V_{E1}}{R_3} = 900\,\mu\text{A}, \qquad I_{C1} \approx I_{E1}$$
4. The current $I_{C1} = 900\,\mu\text{A}$ flows from V_{C1} through R_4: $V_{C1} = (-5) + R_4\,(I_{C1}) = -5 + 3\,\text{k} \times 900\,\mu\text{A} = -2.3\,\text{V}$.
5. Q_2 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E2} = V_{C1} - 0.7 = -3\,\text{V}$.
6. KCL at the emitter of Q_2 (base current neglected):
$$I_{E2} = \frac{V_{E2} - (-5)}{R_6} = 1\,\text{mA}, \qquad I_{C2} \approx I_{E2}$$
7. Q_2 draws $I_{C2} = 1\,\text{mA}$ from V_{C2} through R_5: $V_{C2} = 5 - R_5\,(I_{C2}) = 5 - 3\,\text{k} \times 1\,\text{mA} = 2\,\text{V}$.
8. Q_3 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{E3} = V_{C2} + 0.7 = 2.7\,\text{V}$.
9. KCL at the emitter of Q_3 (base current neglected):
$$I_{E3} = \frac{5 - V_{E3}}{R_7} = 1.28\,\text{mA}, \qquad I_{C3} \approx I_{E3}$$

10. The current $I_{C3} = 1.28\,\text{mA}$ flows from V_{C3} through R_8: $V_{C3} = (-5) + R_8\,(I_{C3}) = -5 + 2.2\,\text{k} \times 1.28\,\text{mA} = -2.19\,\text{V}$.

11. Q_4 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E4} = V_{C3} - 0.7 = -2.89\,\text{V}$.

12. KCL at the emitter of Q_4 (base current neglected):

$$I_{E4} = \frac{V_{E4} - (-5)}{R_9} = 2.11\,\text{mA}, \qquad I_{C4} \approx I_{E4}$$

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 9\,\mu\text{A} \ll$ the 2.5 mA flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 10\,\mu\text{A} \ll$ the 900 μA flowing in the base network ✓
- $I_{B3} = \frac{I_{C3}}{\beta} = 12.8\,\mu\text{A} \ll$ the 1 mA flowing in the base network ✓
- $I_{B4} = \frac{I_{C4}}{\beta} = 21.1\,\mu\text{A} \ll$ the 1.28 mA flowing in the base network ✓
- $V_{EC1} = V_{E1} - V_{C1} = 3.2 - (-2.3) = 5.5\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{CE2} = V_{C2} - V_{E2} = 2 - (-3) = 5\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓
- $V_{EC3} = V_{E3} - V_{C3} = 2.7 - (-2.189) = 4.89\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_3$ is active ✓
- $V_{CE4} = 5 - V_{E4} = 5 - (-2.889) = 7.89\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_4$ is active ✓

Step 4 — How good was $I_B \approx 0$?

If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

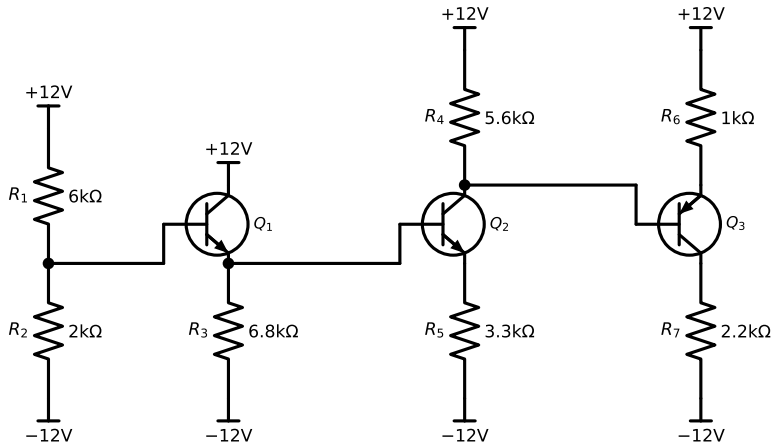
transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	900 μA	888 μA	1.4%
Q_2	1 mA	958 μA	4.4%
Q_3	1.28 mA	1.18 mA	8.7%
Q_4	2.11 mA	1.83 mA	15%

The differences grow along the chain: every base current slightly shifts the collector voltage that biases the next transistor, and $\alpha = \beta/(\beta+1)$ makes each I_C about 1 % smaller than its I_E. A shift is amplified when the next stage has only a small voltage across its emitter resistor. Hand analysis with $I_B \approx 0$ is still the standard first pass.

Answer

$I_{C1} = 900\,\mu\text{A}$, $I_{C2} = 1\,\text{mA}$, $I_{C3} = 1.28\,\text{mA}$, $I_{C4} = 2.11\,\text{mA}$
all transistors active, all base currents negligible.

Solution 4A.15 **Hard**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. No current flows into the base (I_B is negligible), so V_{B1} is set by the divider R_1–R_2 between 12 and (-12) (Lecture-1 tool):
$$V_{B1} = -12 + (12 - (-12))\frac{2\,\text{k}}{6\,\text{k} + 2\,\text{k}} = -6\,\text{V}$$

2. Q_1 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E1} = V_{B1} - 0.7 = -6.7\,\text{V}$.

3. Q_2 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E2} = V_{E1} - 0.7 = -7.4\,\text{V}$.

4. KCL at the emitter of Q_1 (base current neglected):
$$I_{E1} = \frac{V_{E1} - (-12)}{R_3} = 779\,\mu\text{A}, \qquad I_{C1} \approx I_{E1}$$

5. KCL at the emitter of Q_2 (base current neglected):
$$I_{E2} = \frac{V_{E2} - (-12)}{R_5} = 1.39\,\text{mA}, \qquad I_{C2} \approx I_{E2}$$

6. Q_2 draws $I_{C2} = 1.39\,\text{mA}$ from V_{C2} through R_4: $V_{C2} = 12 - R_4\,(I_{C2}) = 12 - 5.6\,\text{k} \times 1.39\,\text{mA} = 4.19\,\text{V}$.

7. Q_3 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{E3} = V_{C2} + 0.7 = 4.89\,\text{V}$.

8. KCL at the emitter of Q_3 (base current neglected):
$$I_{E3} = \frac{12 - V_{E3}}{R_6} = 7.11\,\text{mA}, \qquad I_{C3} \approx I_{E3}$$

9. The current $I_{C3} = 7.11\,\text{mA}$ flows from V_{C3} through R_7: $V_{C3} = (-12) + R_7\,(I_{C3}) = -12 + 2.2\,\text{k} \times 7.11\,\text{mA} = 3.63\,\text{V}$.

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 7.79\,\mu\text{A} \ll$ the 3 mA flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 13.9\,\mu\text{A} \ll$ the 779 μA flowing in the base network ✓
- $I_{B3} = \frac{I_{C3}}{\beta} = 71.1\,\mu\text{A} \ll$ the 1.39 mA flowing in the base network ✓
- $V_{CE1} = 12 - V_{E1} = 12 - (-6.7) = 18.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{CE2} = V_{C2} - V_{E2} = 4.194 - (-7.4) = 11.6\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓
- $V_{EC3} = V_{E3} - V_{C3} = 4.894 - (3.633) = 1.26\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_3$ is active ✓

Step 4 — How good was $I_B \approx 0$?

If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

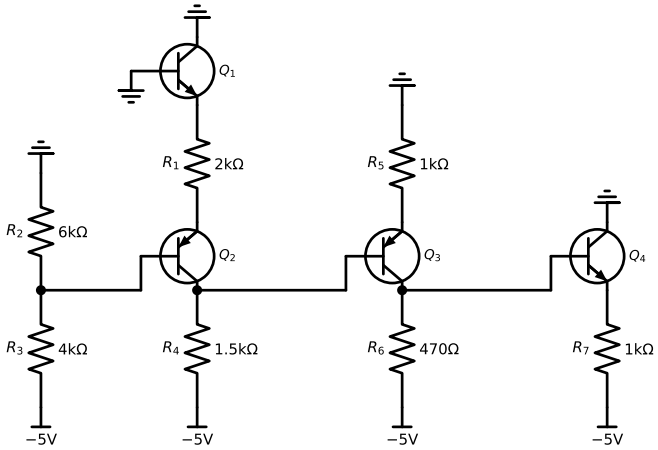
transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	779 μA	784 μA	-0.54%
Q_2	1.39 mA	1.38 mA	1.3%
Q_3	7.11 mA	6.58 mA	8.1%

The differences grow along the chain: every base current slightly shifts the collector voltage that biases the next transistor, and $\alpha = \beta/(\beta+1)$ makes each I_C about 1 % smaller than its I_E. A shift is amplified when the next stage has only a small voltage across its emitter resistor. Hand analysis with $I_B \approx 0$ is still the standard first pass.

Answer

$I_{C1} = 779\,\mu\text{A}, \quad I_{C2} = 1.39\,\text{mA}, \quad I_{C3} = 7.11\,\text{mA}$
all transistors active, all base currents negligible.

Solution 4A.16 **Hard**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. Q_1 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E1} = 0 - 0.7 = -0.7\,\text{V}$.
2. No current flows into the base (I_B is negligible), so V_{B2} is set by the divider R_2–R_3 between 0 and (-5) (Lecture-1 tool):
$$V_{B2} = -5 + (0 - (-5))\frac{4\,\text{k}}{6\,\text{k} + 4\,\text{k}} = -3\,\text{V}$$
3. Q_2 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{E2} = V_{B2} + 0.7 = -2.3\,\text{V}$.
4. KCL at the emitter of Q_1 (base current neglected):
$$I_{E1} = \frac{V_{E1} - V_{E2}}{R_1} = 800\,\mu\text{A}, \qquad I_{C1} \approx I_{E1}$$
5. KCL at the emitter of Q_2 (base current neglected):
$$I_{E2} = \frac{V_{E1} - V_{E2}}{R_1} = 800\,\mu\text{A}, \qquad I_{C2} \approx I_{E2}$$
6. The current $I_{C2} = 800\,\mu\text{A}$ flows from V_{C2} through R_4: $V_{C2} = (-5) + R_4\,(I_{C2}) = -5 + 1.5\,\text{k} \times 800\,\mu\text{A} = -3.8\,\text{V}$.
7. Q_3 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{E3} = V_{C2} + 0.7 = -3.1\,\text{V}$.
8. KCL at the emitter of Q_3 (base current neglected):
$$I_{E3} = \frac{0 - V_{E3}}{R_5} = 3.1\,\text{mA}, \qquad I_{C3} \approx I_{E3}$$
9. The current $I_{C3} = 3.1\,\text{mA}$ flows from V_{C3} through R_6: $V_{C3} = (-5) + R_6\,(I_{C3}) = -5 + 470 \times 3.1\,\text{mA} = -3.54\,\text{V}$.

10. Q_4 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E4} = V_{C3} - 0.7 = -4.24\,\text{V}$.

11. KCL at the emitter of Q_4 (base current neglected):

$$I_{E4} = \frac{V_{E4} - (-5)}{R_7} = 757\,\mu\text{A}, \qquad I_{C4} \approx I_{E4}$$

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 8\,\mu\text{A} \ll$ the 3.1 mA flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 8\,\mu\text{A} \ll$ the 500 μA flowing in the base network ✓
- $I_{B3} = \frac{I_{C3}}{\beta} = 31\,\mu\text{A} \ll$ the 800 μA flowing in the base network ✓
- $I_{B4} = \frac{I_{C4}}{\beta} = 7.57\,\mu\text{A} \ll$ the 3.1 mA flowing in the base network ✓
- $V_{CE1} = 0 - V_{E1} = 0 - (-0.7) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{EC2} = V_{E2} - V_{C2} = -2.3 - (-3.8) = 1.5\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓
- $V_{EC3} = V_{E3} - V_{C3} = -3.1 - (-3.543) = 0.443\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_3$ is active ✓
- $V_{CE4} = 0 - V_{E4} = 0 - (-4.243) = 4.24\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_4$ is active ✓

Step 4 — How good was $I_B \approx 0$?

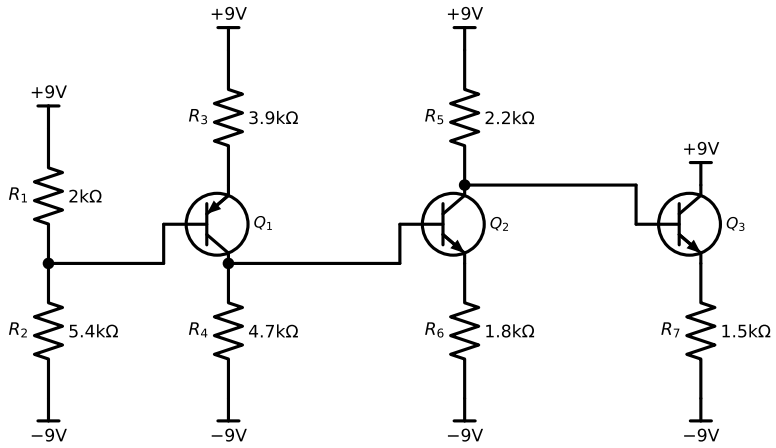
If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	800 μA	783 μA	2.2%
Q_2	800 μA	783 μA	2.2%
Q_3	3.1 mA	3.05 mA	1.7%
Q_4	757 μA	723 μA	4.7%

Answer

$I_{C1} = 800\,\mu\text{A}$, $I_{C2} = 800\,\mu\text{A}$, $I_{C3} = 3.1\,\text{mA}$, $I_{C4} = 757\,\mu\text{A}$
all transistors active, all base currents negligible.

Solution 4A.17 **Hard**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. No current flows into the base (I_B is negligible), so V_{B1} is set by the divider R_1–R_2 between 9 and (-9) (Lecture-1 tool):
$$V_{B1} = -9 + (9 - (-9))\frac{5.4\,\text{k}}{2\,\text{k} + 5.4\,\text{k}} = 4.14\,\text{V}$$

2. Q_1 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{E1} = V_{B1} + 0.7 = 4.84\,\text{V}$.

3. KCL at the emitter of Q_1 (base current neglected):
$$I_{E1} = \frac{9 - V_{E1}}{R_3} = 1.07\,\text{mA}, \qquad I_{C1} \approx I_{E1}$$

4. The current $I_{C1} = 1.07\,\text{mA}$ flows from V_{C1} through R_4: $V_{C1} = (-9) + R_4\,(I_{C1}) = -9 + 4.7\,\text{k} \times 1.07\,\text{mA} = -3.98\,\text{V}$.

5. Q_2 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E2} = V_{C1} - 0.7 = -4.68\,\text{V}$.

6. KCL at the emitter of Q_2 (base current neglected):
$$I_{E2} = \frac{V_{E2} - (-9)}{R_6} = 2.4\,\text{mA}, \qquad I_{C2} \approx I_{E2}$$

7. Q_2 draws $I_{C2} = 2.4\,\text{mA}$ from V_{C2} through R_5: $V_{C2} = 9 - R_5\,(I_{C2}) = 9 - 2.2\,\text{k} \times 2.4\,\text{mA} = 3.72\,\text{V}$.

8. Q_3 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E3} = V_{C2} - 0.7 = 3.02\,\text{V}$.

9. KCL at the emitter of Q_3 (base current neglected):
$$I_{E3} = \frac{V_{E3} - (-9)}{R_7} = 8.01\,\text{mA}, \qquad I_{C3} \approx I_{E3}$$

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 10.7\,\mu\text{A} \ll$ the 2.43 mA flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 24\,\mu\text{A} \ll$ the 1.07 mA flowing in the base network ✓
- $I_{B3} = \frac{I_{C3}}{\beta} = 80.1\,\mu\text{A} \ll$ the 2.4 mA flowing in the base network ✓
- $V_{EC1} = V_{E1} - V_{C1} = 4.835 - (-3.981) = 8.82\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{CE2} = V_{C2} - V_{E2} = 3.721 - (-4.681) = 8.4\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓
- $V_{CE3} = 9 - V_{E3} = 9 - (3.021) = 5.98\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_3$ is active ✓

Step 4 — How good was $I_B \approx 0$?

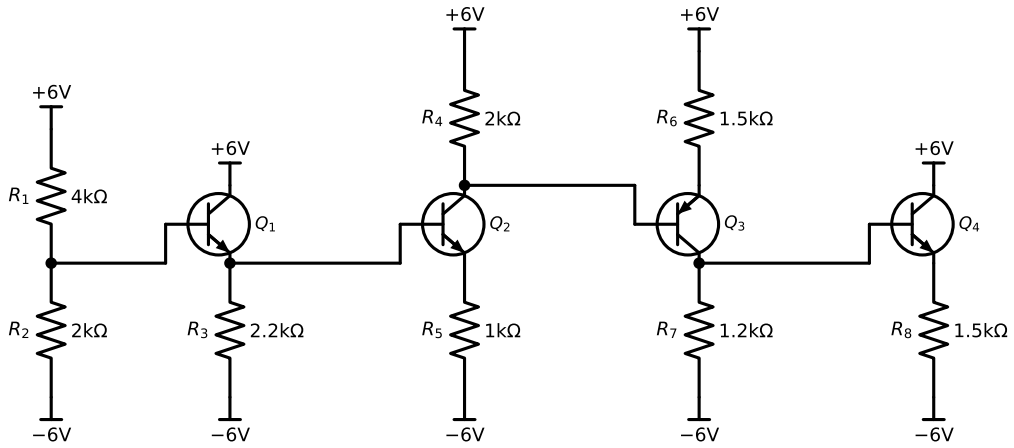
If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	1.07 mA	1.05 mA	1.4%
Q_2	2.4 mA	2.28 mA	5.3%
Q_3	8.01 mA	7.99 mA	0.26%

Answer

$I_{C1} = 1.07\,\text{mA}, \quad I_{C2} = 2.4\,\text{mA}, \quad I_{C3} = 8.01\,\text{mA}$
all transistors active, all base currents negligible.

Solution 4A.18 **Hard**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. No current flows into the base (I_B is negligible), so V_{B1} is set by the divider R_1–R_2 between 6 and (-6) (Lecture-1 tool):
$$V_{B1} = -6 + (6 - (-6))\frac{2\,\text{k}}{4\,\text{k} + 2\,\text{k}} = -2\,\text{V}$$
2. Q_1 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E1} = V_{B1} - 0.7 = -2.7\,\text{V}$.
3. Q_2 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E2} = V_{E1} - 0.7 = -3.4\,\text{V}$.
4. KCL at the emitter of Q_1 (base current neglected):
$$I_{E1} = \frac{V_{E1} - (-6)}{R_3} = 1.5\,\text{mA}, \qquad I_{C1} \approx I_{E1}$$
5. KCL at the emitter of Q_2 (base current neglected):
$$I_{E2} = \frac{V_{E2} - (-6)}{R_5} = 2.6\,\text{mA}, \qquad I_{C2} \approx I_{E2}$$
6. Q_2 draws $I_{C2} = 2.6\,\text{mA}$ from V_{C2} through R_4: $V_{C2} = 6 - R_4\,(I_{C2}) = 6 - 2\,\text{k} \times 2.6\,\text{mA} = 0.8\,\text{V}$.
7. Q_3 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{E3} = V_{C2} + 0.7 = 1.5\,\text{V}$.
8. KCL at the emitter of Q_3 (base current neglected):
$$I_{E3} = \frac{6 - V_{E3}}{R_6} = 3\,\text{mA}, \qquad I_{C3} \approx I_{E3}$$
9. The current $I_{C3} = 3\,\text{mA}$ flows from V_{C3} through R_7: $V_{C3} = (-6) + R_7\,(I_{C3}) = -6 + 1.2\,\text{k} \times 3\,\text{mA} = -2.4\,\text{V}$.

10. Q_4 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E4} = V_{C3} - 0.7 = -3.1\,\text{V}$.

11. KCL at the emitter of Q_4 (base current neglected):

$$I_{E4} = \frac{V_{E4} - (-6)}{R_8} = 1.93\,\text{mA}, \qquad I_{C4} \approx I_{E4}$$

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 15\,\mu\text{A} \ll$ the 2 mA flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 26\,\mu\text{A} \ll$ the 1.5 mA flowing in the base network ✓
- $I_{B3} = \frac{I_{C3}}{\beta} = 30\,\mu\text{A} \ll$ the 2.6 mA flowing in the base network ✓
- $I_{B4} = \frac{I_{C4}}{\beta} = 19.3\,\mu\text{A} \ll$ the 3 mA flowing in the base network ✓
- $V_{CE1} = 6 - V_{E1} = 6 - (-2.7) = 8.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{CE2} = V_{C2} - V_{E2} = 0.8 - (-3.4) = 4.2\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓
- $V_{EC3} = V_{E3} - V_{C3} = 1.5 - (-2.4) = 3.9\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_3$ is active ✓
- $V_{CE4} = 6 - V_{E4} = 6 - (-3.1) = 9.1\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_4$ is active ✓

Step 4 — How good was $I_B \approx 0$?

If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	1.5 mA	1.5 mA	−0.095%
Q_2	2.6 mA	2.55 mA	1.8%
Q_3	3 mA	2.87 mA	4.4%
Q_4	1.93 mA	1.8 mA	7.5%

The differences grow along the chain: every base current slightly shifts the collector voltage that biases the next transistor, and $\alpha = \beta/(\beta+1)$ makes each I_C about 1 % smaller than its I_E. A shift is amplified when the next stage has only a small voltage across its emitter resistor. Hand analysis with $I_B \approx 0$ is still the standard first pass.

Answer

$I_{C1} = 1.5\,\text{mA}, \quad I_{C2} = 2.6\,\text{mA}, \quad I_{C3} = 3\,\text{mA}, \quad I_{C4} = 1.93\,\text{mA}$
all transistors active, all base currents negligible.